

Airport Management Information System

CMPE343 Term Project - Relational Database Design and SQL Implementation

Ercan Airport Management Database System

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Project Overview and ERD Context

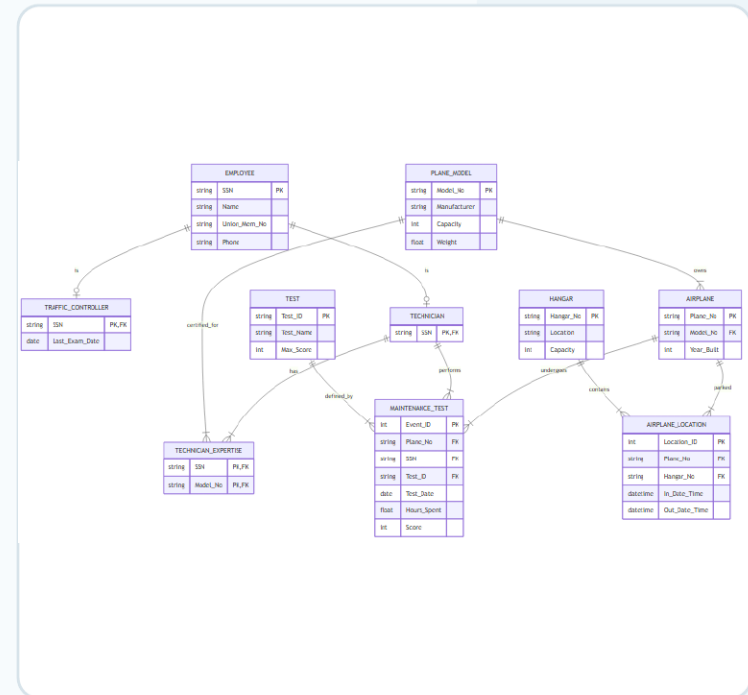
The first design step was to model the airport domain as connected entities and relationships.

System Scope

- The database represents an airport management environment focused on aircraft, staff, hangars, and maintenance operations.
- The ERD translates the real-world aviation workflow into relational tables that can be implemented using SQL.

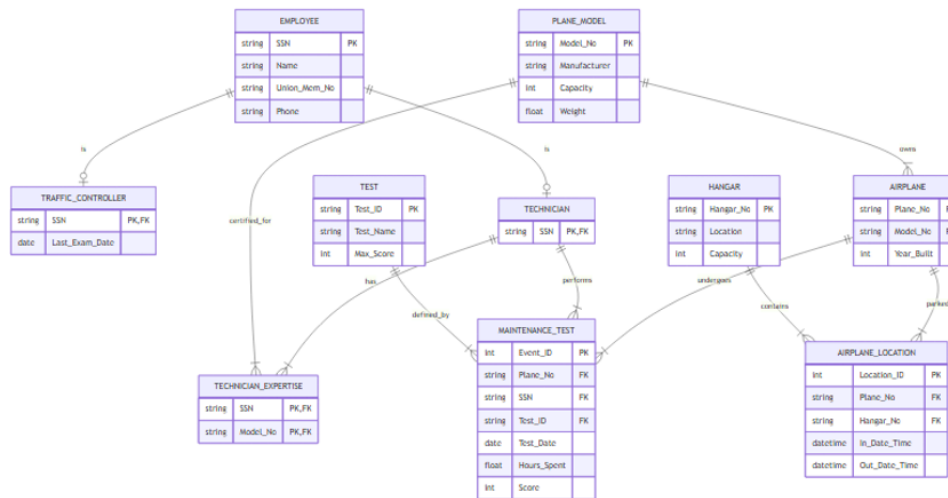
Main ERD Components

- Employee is the parent entity for staff roles such as Technician and Traffic Controller.
- Airplane and Plane_Model describe aircraft identity and technical capacity.
- Maintenance_Test connects airplanes, technicians, tests, scores, dates, and working hours.
- Airplane_Location records which hangar stores each airplane and when it enters or leaves.



Entity-Relationship Design

Entity Relationship (ER) Diagram



Relationship Logic

- One plane model can define many airplanes.
- One airplane can have many maintenance test records.
- One technician can be certified for multiple plane models.
- Hangar usage is tracked historically through location records.

Database Meaning

- The diagram provides the blueprint for primary keys, foreign keys, and junction tables before SQL implementation.

System Objectives

Structural Integrity

Develop a normalized relational model that minimizes unnecessary duplication and keeps airport data consistent.

SQL Implementation

Create the schema with DDL commands and populate it with realistic DML sample records.

Operational Tracking

Monitor hangar occupancy, aircraft information, employee roles, and maintenance history.

Analytical Queries

Use relational and statistical SELECT queries to extract meaningful operational results.

Key Design Assumptions

Employee Hierarchy

- Employee stores shared staff details such as SSN, name, union membership number, and phone.
- Technician and Traffic_Controller are role-specific extensions connected through SSN.

Aircraft Model Logic

- Each airplane belongs to a single Plane_Model.
- A Plane_Model can describe many airplanes with the same manufacturer, capacity, and weight structure.

Maintenance Tracking

- Every maintenance event records the airplane, technician, test type, date, hours spent, and score.
- This supports both safety compliance and performance analysis.

Temporal Location History

- Airplane_Location uses In_Date_Time and Out_Date_Time to track hangar residency.
- A NULL Out_Date_Time means the aircraft is currently stored in that hangar.

Schema Architecture

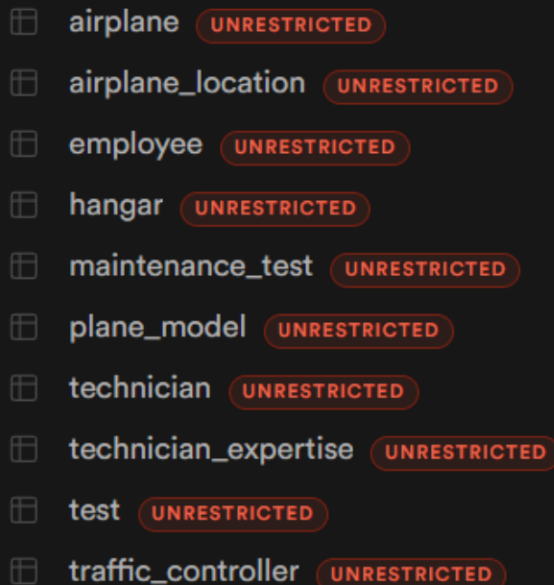
The schema is organized into personnel, fleet, infrastructure, and maintenance groups.

Category	Tables	Main Key	Purpose
Personnel	Employee, Technician, Traffic_Controller	SSN	Stores staff information and separates general emp
Fleet	Plane_Model, Airplane	Model_No, Plane_No	Connects each aircraft to technical model specific
Infrastructure	Hangar, Airplane_Location	Hangar_No, Location_ID	Tracks storage capacity and historical hangar occu
Maintenance	Test, Maintenance_Test	Test_ID, Event_ID	Logs inspection types, technicians, aircraft, scor

After this design foundation, the presentation moves into the SQL setup: DDL table creation, DML data insertion, validation views, and the 15 required query outputs.

SQL Setup 1: DDL - Core Personnel Tables

The first SQL implementation step creates the core staff structure and role-based subtype tables.



- airplane UNRESTRICTED
- airplane_location UNRESTRICTED
- employee UNRESTRICTED
- hangar UNRESTRICTED
- maintenance_test UNRESTRICTED
- plane_model UNRESTRICTED
- technician UNRESTRICTED
- technician_expertise UNRESTRICTED
- test UNRESTRICTED
- traffic_controller UNRESTRICTED

```
CREATE TABLE Employee (  
    SSN VARCHAR(15) PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL,  
    Union_Mem_No VARCHAR(20) UNIQUE,  
    Phone VARCHAR(15)  
);  
  
CREATE TABLE Traffic_Controller (  
    SSN VARCHAR(15) PRIMARY KEY,  
    Last_Exam_Date DATE NOT NULL,  
    FOREIGN KEY (SSN) REFERENCES Employee(SSN) ON DELETE CASCADE  
);  
  
CREATE TABLE Technician (  
    SSN VARCHAR(15) PRIMARY KEY,  
    FOREIGN KEY (SSN) REFERENCES Employee(SSN) ON DELETE CASCADE  
);  
  
CREATE TABLE Plane_Model (  
    Model_No VARCHAR(20) PRIMARY KEY,  
    Manufacturer VARCHAR(50),  
    Capacity INT,  
    Weight DECIMAL(10,2)  
);
```

What this proves

The schema begins with employee-related tables. Primary keys identify each record, while foreign keys connect role-specific entities to the shared employee table.

e shared

SQL Setup 2: DDL - Fleet and Infrastructure

After staff tables, the database creates airplane, model, hangar, and location tables.

```
CREATE TABLE Technician_Expertise (  
    SSN VARCHAR(15),  
    Model_No VARCHAR(20),  
    PRIMARY KEY (SSN, Model_No),  
    FOREIGN KEY (SSN) REFERENCES Technician(SSN) ON DELETE CASCADE,  
    FOREIGN KEY (Model_No) REFERENCES Plane_Model(Model_No) ON DELETE CASCADE  
);  
  
CREATE TABLE Airplane (  
    Plane_No VARCHAR(20) PRIMARY KEY,  
    Model_No VARCHAR(20),  
    Year_Built INT,  
    FOREIGN KEY (Model_No) REFERENCES Plane_Model(Model_No)  
);  
  
CREATE TABLE Hangar (  
    Hangar_No VARCHAR(10) PRIMARY KEY,  
    Location VARCHAR(100),  
    Capacity INT  
);
```

```
CREATE TABLE Airplane_Location (  
    Location_ID INT PRIMARY KEY,  
    Plane_No VARCHAR(20),  
    Hangar_No VARCHAR(10),  
    In_Date_Time TIMESTAMP,  
    Out_Date_Time TIMESTAMP,  
    FOREIGN KEY (Plane_No) REFERENCES Airplane(Plane_No) ON DELETE CASCADE,  
    FOREIGN KEY (Hangar_No) REFERENCES Hangar(Hangar_No) ON DELETE CASCADE  
);  
  
CREATE TABLE Test (  
    Test_ID VARCHAR(10) PRIMARY KEY,  
    Test_Name VARCHAR(100) NOT NULL,  
    Max_Score INT  
);
```

Purpose

- Defines airplane identity and plane model structure.
- Creates hangar storage and location history tables.
- Uses foreign keys to connect aircraft to models and hangars.

```
CREATE TABLE Maintenance_Test (  
    Event_ID INT PRIMARY KEY,  
    Plane_No VARCHAR(20),  
    SSN VARCHAR(15),  
    Test_ID VARCHAR(10),  
    Test_Date DATE NOT NULL,  
    Hours_Spent DECIMAL(5,2),  
    Score INT,  
    FOREIGN KEY (Plane_No) REFERENCES Airplane(Plane_No) ON DELETE CASCADE,  
    FOREIGN KEY (SSN) REFERENCES Technician(SSN),  
    FOREIGN KEY (Test_ID) REFERENCES Test(Test_ID)  
);
```


SQL Setup 3: DML - Core Data Insertion

Once tables are created, INSERT statements populate the schema with sample airport data.

```
-- Insert Employees
INSERT INTO Employee (SSN, Name, Union_Mem_No, Phone) VALUES ('11111', 'Ahmet Yilmaz', 'U001', '555-0101');
INSERT INTO Employee (SSN, Name, Union_Mem_No, Phone) VALUES ('22222', 'Ayse Demir', 'U002', '555-0202');
INSERT INTO Employee (SSN, Name, Union_Mem_No, Phone) VALUES ('33333', 'Mehmet Kaya', 'U003', '555-0303');
INSERT INTO Employee (SSN, Name, Union_Mem_No, Phone) VALUES ('44444', 'Fatma Celik', 'U004', '555-0404');
INSERT INTO Employee (SSN, Name, Union_Mem_No, Phone) VALUES ('55555', 'Ali Tekin', 'U005', '555-0505');

-- Insert Subtypes
INSERT INTO Traffic_Controller (SSN, Last_Exam_Date) VALUES ('11111', '2025-10-15');
INSERT INTO Traffic_Controller (SSN, Last_Exam_Date) VALUES ('22222', '2026-01-20');
INSERT INTO Technician (SSN) VALUES ('33333');
INSERT INTO Technician (SSN) VALUES ('44444');

-- Insert Plane Models
INSERT INTO Plane_Model (Model_No, Manufacturer, Capacity, Weight) VALUES ('B737', 'Boeing', 200, 41000);
INSERT INTO Plane_Model (Model_No, Manufacturer, Capacity, Weight) VALUES ('A320', 'Airbus', 180, 42600);
INSERT INTO Plane_Model (Model_No, Manufacturer, Capacity, Weight) VALUES ('B777', 'Boeing', 350, 134000);

-- Insert Technician Expertise
INSERT INTO Technician_Expertise (SSN, Model_No) VALUES ('33333', 'B737');
INSERT INTO Technician_Expertise (SSN, Model_No) VALUES ('33333', 'A320');
INSERT INTO Technician_Expertise (SSN, Model_No) VALUES ('44444', 'B777');
```

Inserted Records

- Employees and staff subtypes
- Plane models
- Technician expertise links
- Sample values for testing joins and role logic

SQL Setup 4: DML - Aircraft, Hangars and Locations

The next insertion step adds aircraft, storage areas, and hangar movement records.

```
-- Insert Airplanes
INSERT INTO Airplane (Plane_No, Model_No, Year_Built) VALUES ('TC-JAA', 'B737', 2015);
INSERT INTO Airplane (Plane_No, Model_No, Year_Built) VALUES ('TC-JAB', 'A320', 2018);
INSERT INTO Airplane (Plane_No, Model_No, Year_Built) VALUES ('TC-JAC', 'B777', 2020);

-- Insert Hangars
INSERT INTO Hangar (Hangar_No, Location, Capacity) VALUES ('H1', 'North Wing', 5);
INSERT INTO Hangar (Hangar_No, Location, Capacity) VALUES ('H2', 'South Wing', 3);

-- Insert Airplane Locations
INSERT INTO Airplane_Location (Location_ID, Plane_No, Hangar_No, In_Date_Time, Out_Date_Time)
VALUES (1, 'TC-JAA', 'H1', '2026-05-01 10:00:00', '2026-05-02 12:00:00');
INSERT INTO Airplane_Location (Location_ID, Plane_No, Hangar_No, In_Date_Time, Out_Date_Time)
VALUES (2, 'TC-JAB', 'H2', '2026-05-10 08:00:00', NULL);

-- Insert Tests
INSERT INTO Test (Test_ID, Test_Name, Max_Score) VALUES ('T1', 'Engine Diagnostic', 100);
INSERT INTO Test (Test_ID, Test_Name, Max_Score) VALUES ('T2', 'Landing Gear Check', 100);
INSERT INTO Test (Test_ID, Test_Name, Max_Score) VALUES ('T3', 'Avionics System Test', 100);
```

Why it matters

- Airplanes are linked to their model numbers.
- Hangars provide capacity and location details.
- Airplane_Location stores IN/OUT timestamps for historical tracking.

SQL Setup 5: DML - Maintenance Test Records

Maintenance data enables the analytical queries used later in the presentation.

```
-- Insert Maintenance Tests
INSERT INTO Maintenance_Test (Event_ID, Plane_No, SSN, Test_ID, Test_Date, Hours_Spent, Score)
VALUES (1, 'TC-JAA', '33333', 'T1', '2026-05-01', 4.5, 95);
INSERT INTO Maintenance_Test (Event_ID, Plane_No, SSN, Test_ID, Test_Date, Hours_Spent, Score)
VALUES (2, 'TC-JAB', '33333', 'T2', '2026-05-11', 2.0, 98);
INSERT INTO Maintenance_Test (Event_ID, Plane_No, SSN, Test_ID, Test_Date, Hours_Spent, Score)
VALUES (3, 'TC-JAC', '44444', 'T3', '2026-05-12', 6.0, 100);
INSERT INTO Maintenance_Test (Event_ID, Plane_No, SSN, Test_ID, Test_Date, Hours_Spent, Score)
VALUES (4, 'TC-JAA', '33333', 'T2', '2026-05-05', 3.0, 85);
```

Maintenance Data Includes

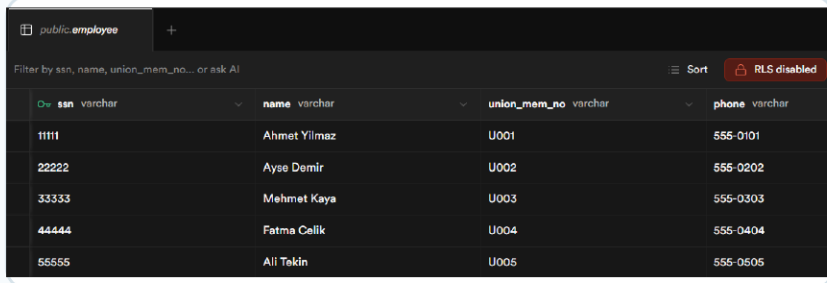
- The airplane being tested
- The technician responsible for the test
- Test date, hours spent, and score

Used For

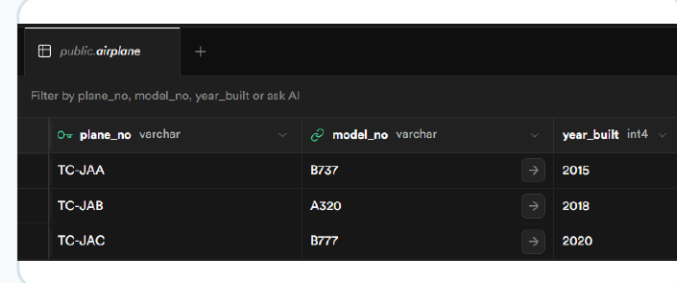
- Average score analysis
- Technician workload measurement
- Most recent test retrieval
- Model-based maintenance reporting

Data Validation: Main Table Views

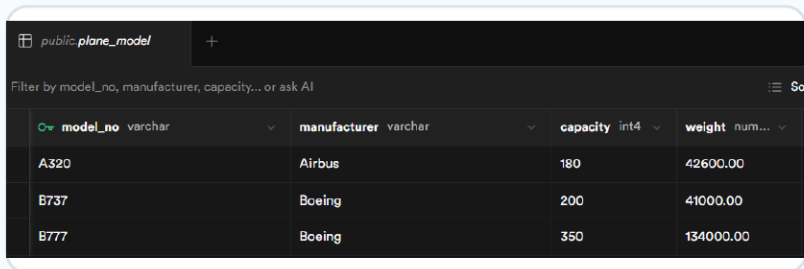
Table views confirm that the inserted data exists in the database and follows the intended structure.



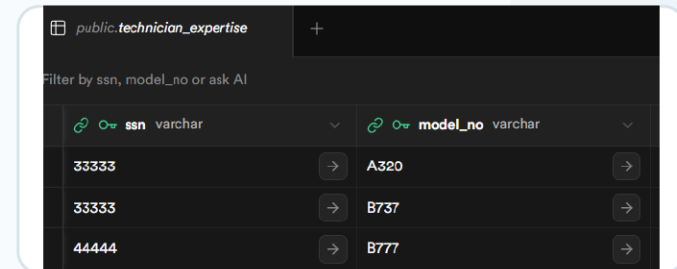
ssn	name	union_mem_no	phone
11111	Ahmet Yilmaz	U001	555-0101
22222	Ayşe Demir	U002	555-0202
33333	Mehmet Kaya	U003	555-0303
44444	Fatma Çelik	U004	555-0404
55555	Ali Tekin	U005	555-0505



plane_no	model_no	year_built
TC-JAA	B737	2015
TC-JAB	A320	2018
TC-JAC	B777	2020



model_no	manufacturer	capacity	weight
A320	Airbus	180	42600.00
B737	Boeing	200	41000.00
B777	Boeing	350	134000.00



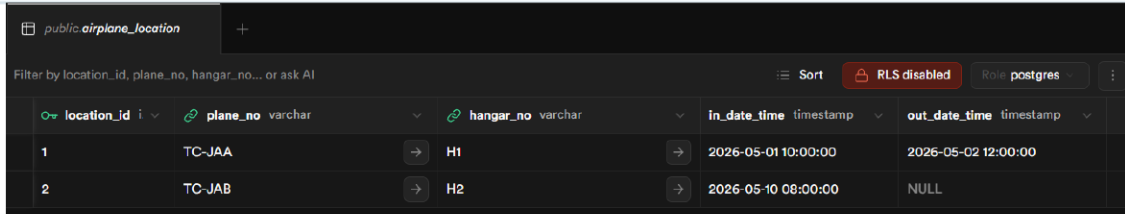
ssn	model_no
33333	A320
33333	B737
44444	B777

Validation Result

The main table views show that personnel, aircraft, plane models, and technician expertise records were inserted successfully.

Data Validation: Operational Table Views

Operational tables prove that hangar usage and maintenance records can be tracked across the system.

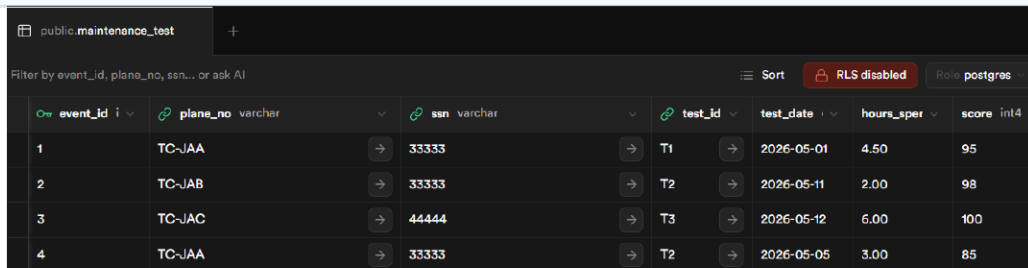


public.airplane_location

Filter by location_id, plane_no, hangar_no... or ask AI

Sort RLS disabled Role postgres

location_id	plane_no	hangar_no	In_date_time	out_date_time
1	TC-JAA	H1	2026-05-01 10:00:00	2026-05-02 12:00:00
2	TC-JAB	H2	2026-05-10 08:00:00	NULL



public.maintenance_test

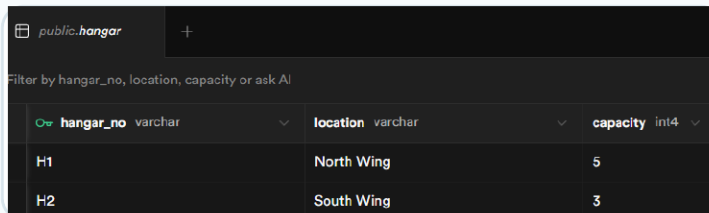
Filter by event_id, plane_no, ssn... or ask AI

Sort RLS disabled Role postgres

event_id	plane_no	ssn	test_id	test_date	hours_sper	score
1	TC-JAA	33333	T1	2026-05-01	4.50	95
2	TC-JAB	33333	T2	2026-05-11	2.00	98
3	TC-JAC	44444	T3	2026-05-12	6.00	100
4	TC-JAA	33333	T2	2026-05-05	3.00	85

Purpose

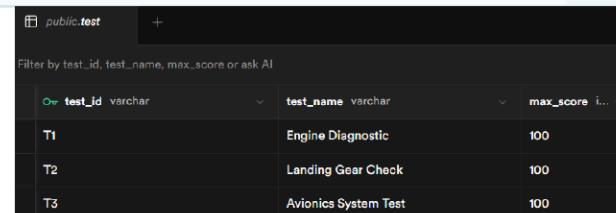
- Location records support current hangar tracking.
- Maintenance records support score, date, and workload analysis.
- Test definitions standardize inspection categories.



public.hangar

Filter by hangar_no, location, capacity or ask AI

hangar_no	location	capacity
H1	North Wing	5
H2	South Wing	3



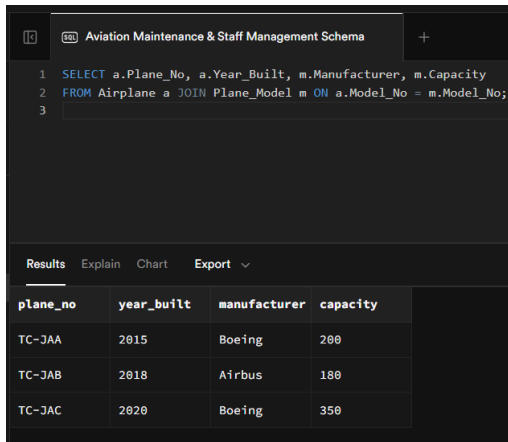
public.test

Filter by test_id, test_name, max_score or ask AI

test_id	test_name	max_score
T1	Engine Diagnostic	100
T2	Landing Gear Check	100
T3	Avionics System Test	100

Query Analysis: Q1-Q3

Basic Retrieval and Grouping



The screenshot shows a SQL query in a dark-themed editor. The query is: `1 SELECT a.Plane_No, a.Year_Built, m.Manufacturer, m.Capacity`
`2 FROM Airplane a JOIN Plane_Model m ON a.Model_No = m.Model_No;`
`3`

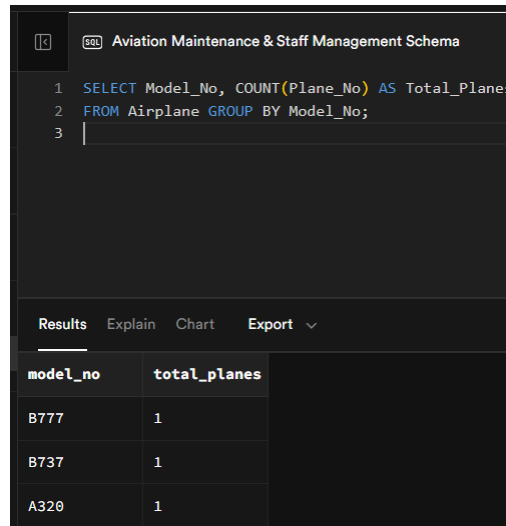
Below the query, the results are displayed in a table with columns: plane_no, year_built, manufacturer, capacity. The data rows are: TC-JAA (2015, Boeing, 200), TC-JAB (2018, Airbus, 180), and TC-JAC (2020, Boeing, 350).

plane_no	year_built	manufacturer	capacity
TC-JAA	2015	Boeing	200
TC-JAB	2018	Airbus	180
TC-JAC	2020	Boeing	350

Q1 - Airplanes with model capacity

Purpose

Uses JOIN to combine Airplane and Plane_Model, showing each aircraft with its manufacturer and capacity.



The screenshot shows a SQL query in a dark-themed editor. The query is: `1 SELECT Model_No, COUNT(Plane_No) AS Total_Planes`
`2 FROM Airplane GROUP BY Model_No;`
`3`

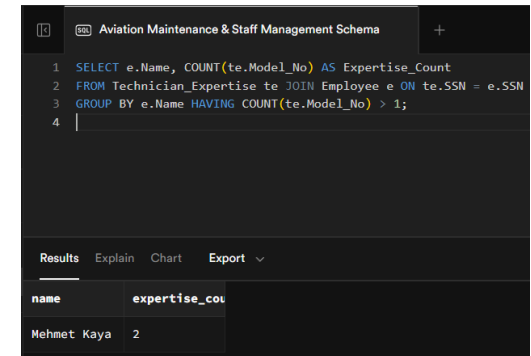
Below the query, the results are displayed in a table with columns: model_no, total_planes. The data rows are: B777 (1), B737 (1), and A320 (1).

model_no	total_planes
B777	1
B737	1
A320	1

Q2 - Airplanes per model

Purpose

Uses GROUP BY and COUNT to show how many aircraft belong to each plane model.



The screenshot shows a SQL query in a dark-themed editor. The query is: `1 SELECT e.Name, COUNT(te.Model_No) AS Expertise_Count`
`2 FROM Technician_Expertise te JOIN Employee e ON te.SSN = e.SSN`
`3 GROUP BY e.Name HAVING COUNT(te.Model_No) > 1;`
`4`

Below the query, the results are displayed in a table with columns: name, expertise_cou. The data row is: Mehmet Kaya (2).

name	expertise_cou
Mehmet Kaya	2

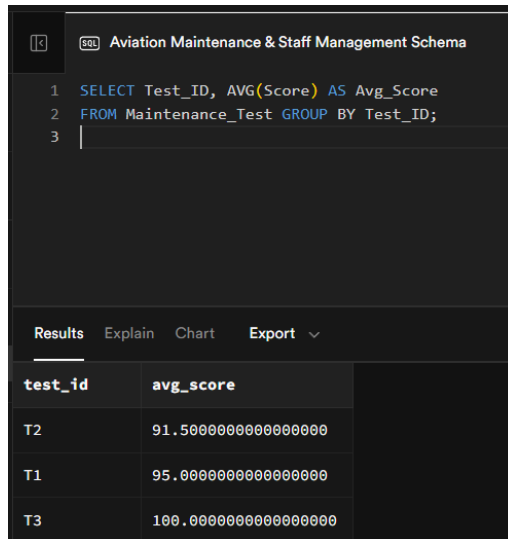
Q3 - Technicians with multiple expertise

Purpose

Finds technicians qualified for more than one aircraft model using grouping and HAVING.

Query Analysis: Q4-Q6

Aggregates and Hangar Tracking



The screenshot shows a SQL query editor with the following query:

```
1 SELECT Test_ID, AVG(Score) AS Avg_Score
2 FROM Maintenance_Test GROUP BY Test_ID;
3
```

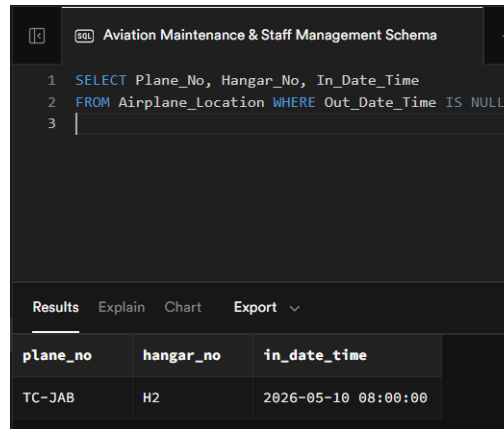
Below the query, the results are displayed in a table with two columns: **test_id** and **avg_score**. The results are as follows:

test_id	avg_score
T2	91.50000000000000
T1	95.00000000000000
T3	100.00000000000000

Q4 - Average test score by type

Purpose

Calculates the average maintenance score for each test type.



The screenshot shows a SQL query editor with the following query:

```
1 SELECT Plane_No, Hangar_No, In_Date_Time
2 FROM Airplane_Location WHERE Out_Date_Time IS NULL;
3
```

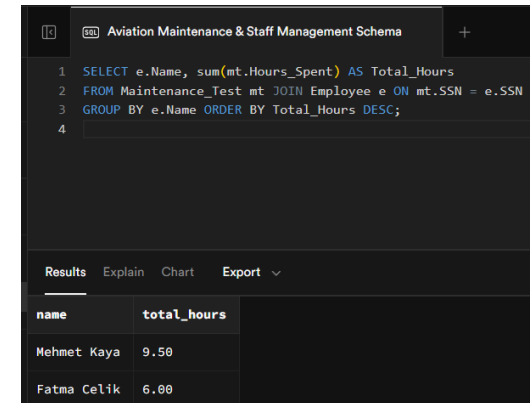
Below the query, the results are displayed in a table with three columns: **plane_no**, **hangar_no**, and **in_date_time**. The results are as follows:

plane_no	hangar_no	in_date_time
TC-JAB	H2	2026-05-10 08:00:00

Q5 - Currently parked airplanes

Purpose

Filters Airplane_Location records where Out_Date_Time is NULL to find current hangar occupancy.



The screenshot shows a SQL query editor with the following query:

```
1 SELECT e.Name, sum(mt.Hours_Spent) AS Total_Hours
2 FROM Maintenance_Test mt JOIN Employee e ON mt.SSN = e.SSN
3 GROUP BY e.Name ORDER BY Total_Hours DESC;
4
```

Below the query, the results are displayed in a table with two columns: **name** and **total_hours**. The results are as follows:

name	total_hours
Mehmet Kaya	9.50
Fatma Celik	6.00

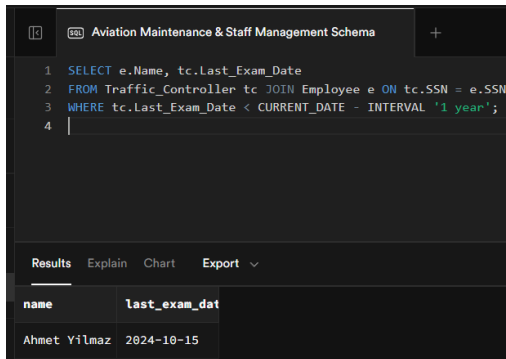
Q6 - Technician workload

Purpose

Sums Hours_Spent per technician and orders the result by workload.

Query Analysis: Q7-Q9

Date, Month and Score Filters



The screenshot shows a SQL query editor with the following query:

```
1 SELECT e.Name, tc.Last_Exam_Date
2 FROM Traffic_Controller tc JOIN Employee e ON tc.SSN = e.SSN
3 WHERE tc.Last_Exam_Date < CURRENT_DATE - INTERVAL '1 year';
4
```

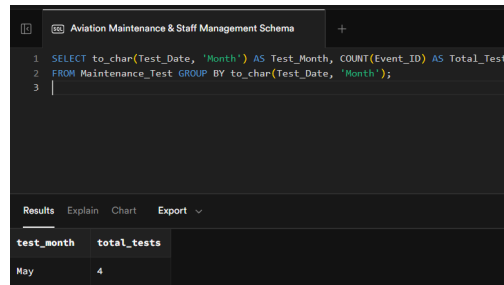
Below the query, the results are displayed in a table with columns 'name' and 'last_exam_dat'.

name	last_exam_dat
Ahmet Yilmaz	2024-10-15

Q7 - Expired medical exams

Purpose

Identifies traffic controllers whose last medical exam is older than one year.



The screenshot shows a SQL query editor with the following query:

```
1 SELECT to_char(Test_Date, 'Month') AS Test_Month, COUNT(Event_ID) AS Total_Tests
2 FROM Maintenance_Test GROUP BY to_char(Test_Date, 'Month');
3
```

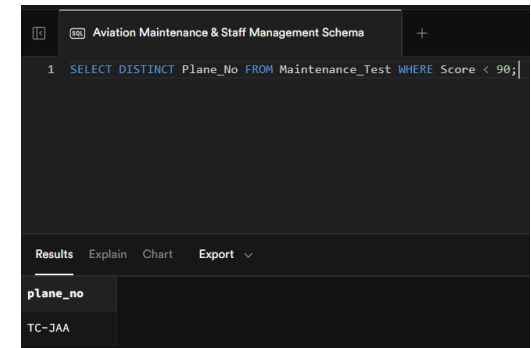
Below the query, the results are displayed in a table with columns 'test_month' and 'total_tests'.

test_month	total_tests
May	4

Q8 - Tests per month

Purpose

Extracts the month from Test_Date and counts maintenance tests per month.



The screenshot shows a SQL query editor with the following query:

```
1 SELECT DISTINCT Plane_No FROM Maintenance_Test WHERE Score < 90;
```

Below the query, the results are displayed in a table with column 'plane_no'.

plane_no
TC-JAA

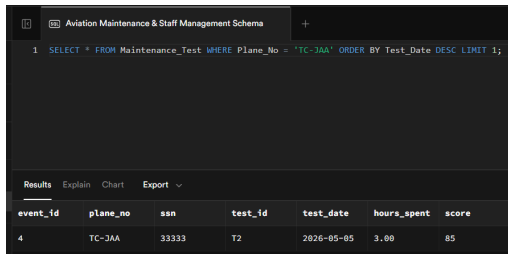
Q9 - Low-scoring airplanes

Purpose

Finds aircraft that scored below 90 in at least one maintenance test.

Query Analysis: Q10-Q12

Recent Events, Roles and Capacity



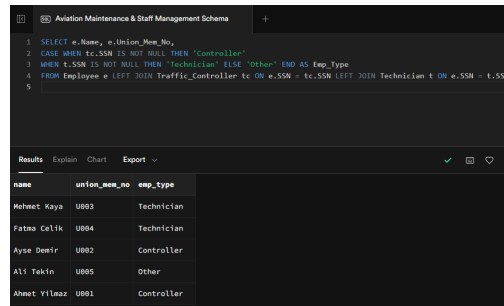
The screenshot shows a SQL query in a dark-themed editor: `1 SELECT * FROM Maintenance_Test WHERE Plane_No = 'TC-JAA' ORDER BY Test_Date DESC LIMIT 1;`. Below the query, the results are displayed in a table with columns: event_id, plane_no, ssn, test_id, test_date, hours_spent, and score. The results table contains one row with the following values: 4, TC-JAA, 33333, T2, 2026-05-05, 3.00, and 85.

event_id	plane_no	ssn	test_id	test_date	hours_spent	score
4	TC-JAA	33333	T2	2026-05-05	3.00	85

Q10 - Most recent test for TC-JAA

Purpose

Retrieves the latest maintenance test record for a specific airplane.



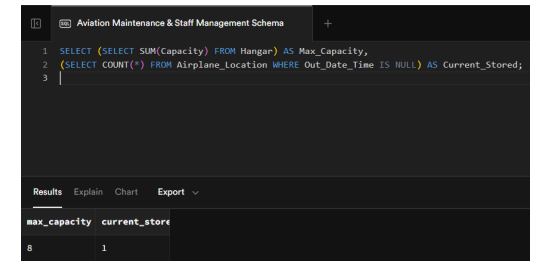
The screenshot shows a SQL query in a dark-themed editor: `1 SELECT e.Name, e.Union_No, e.Type
2 CASE WHEN tc.SSN IS NOT NULL THEN 'Controller'
3 WHEN t.SSN IS NOT NULL THEN 'Technician' ELSE 'Other' END AS Emp_Type
4 FROM Employee e LEFT JOIN Traffic_Controller tc ON e.SSN = tc.SSN LEFT JOIN Technician t ON e.SSN = t.SSN;
5`. Below the query, the results are displayed in a table with columns: name, union_no, and emp_type. The results table contains five rows with the following values: Mehmet Kaya, U003, Technician; Fatma Celik, U004, Technician; Ayse Demir, U002, Controller; Ali Tekin, U005, Other; and Ahmet Yilmaz, U001, Controller.

name	union_no	emp_type
Mehmet Kaya	U003	Technician
Fatma Celik	U004	Technician
Ayşe Demir	U002	Controller
Ali Tekin	U005	Other
Ahmet Yılmaz	U001	Controller

Q11 - Employee role classification

Purpose

Uses outer joins and CASE logic to label employees as controller, technician, or general staff.



The screenshot shows a SQL query in a dark-themed editor: `1 SELECT (SELECT SUM(Capacity) FROM Hangar) AS Max_Capacity,
2 (SELECT COUNT(*) FROM Airplane_Location WHERE Out_Date_Time IS NULL) AS Current_Stored;
3`. Below the query, the results are displayed in a table with columns: max_capacity and current_stored. The results table contains one row with the following values: 8 and 1.

max_capacity	current_stored
8	1

Q12 - Capacity vs current aircraft

Purpose

Compares total hangar capacity with the number of currently stored aircraft.

Query Analysis: Q13-Q15

Advanced Subqueries and Final Statistics

```
1 SELECT Plane_No FROM Airplane
2 WHERE Plane_No NOT IN (SELECT DISTINCT Plane_No FROM Maintenance_Test);
3
```

plane_no
TC-3AD

Q13 - Aircraft without maintenance

Purpose

Uses a subquery to find aircraft with no maintenance records.

```
1 SELECT e.Name, mt.Score FROM Maintenance_Test mt
2 JOIN Employee e ON mt.SSN = e.SSN
3 WHERE mt.Score = (SELECT MAX(Score) FROM Maintenance_Test);
4
```

name	score
Fatma Celik	100

Q14 - Highest test score technician

Purpose

Finds the technician associated with the highest recorded maintenance score.

```
1 SELECT a.Model_No, COUNT(mt.Event_ID) AS Total_Tests, ROUND(AVG(mt.Hours_Spent),
2 ) AS Avg_Hours
3 FROM Maintenance_Test mt JOIN Airplane a ON mt.Plane_No = a.Plane_No GROUP BY a.Model_No;
4
```

model_no	total_tests	avg_hours
B777	1	6.00
B737	2	3.75
A320	1	2.00

Q15 - Model-based maintenance statistics

Purpose

Groups maintenance records by aircraft model and calculates total tests and average hours.

Integrity and Normalization

Third Normal Form

The database separates entities so that non-key attributes depend on the correct primary key. This reduces update, insertion, and deletion anomalies.

Foreign Key Consistency

Relationships between airplanes, models, employees, technicians, tests, and hangars are enforced through foreign keys.

Many-to-Many Resolution

Technician_Expertise acts as a junction table between technicians and plane models, making certifications easier to query.

Temporal Tracking

Airplane_Location records historical hangar movements and supports current occupancy checks by using NULL Out_Date_Time values.

Conclusion and Future Enhancements

Conclusion

- The project successfully models airport operations through a relational database.
- It includes ER design, DDL table creation, DML insertion, table validation, and 15 SQL query outputs.
- The system supports aircraft tracking, maintenance reporting, employee role management, and hangar monitoring.

Future Enhancements

- Develop a web interface for airport staff.
- Add automated alerts for upcoming maintenance tests.
- Include role-based access control.
- Create dashboard charts for hangar capacity and maintenance performance.



Thank You

Questions or technical discussion?

CMPE343 - Airport Management Information System

Cyprus International University | Instructor: Yasemin Bay